

ActInf Textbook Group ~ Cohort 8 ~ Session 7

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All right, welcome back to cohort 8. So we are in the first discussion of chapter 4. Where would anyone like to begin? Could be any section of four or any any question or where four fits in or just anywhere anyone wants to be in first. Yes, Sophia. Um, I I found this chapter quite heavy in math and I'm I'm still working through it. I'm just wondering like how much of the math is required to know for like modeling or for being able to go through papers in the field of inference. I I suppose we don't have to like know everything by heart but like which figures or equations do you think would be very essential to go through to be like the foundation for understanding at inference? No. What does anyone think on that? I'm sorry, I was looking through the textbook. Um, seems like a question for I I I think it's a super interesting question. One piece that comes to mind is like how much math would be required for doing uh neuroimaging like SPM modeling or just any other kind of statistical using a linear regression in Google Sheets or Excel or R or Python or using uh uh some sort of statistics. It's like there's a lot of math that that stems from and describes and kind of gives rise to those statistics equations and and same could be said um for going into the like you know what gives rise to the linear algebra or the probabilistic systems part. So it's in it it's like this this is a really a critical and kind of that that is the question like we get to chapter 4 and it's like okay this is where the the two roads came in from both sides and now this is where it actually is going to be all right this is the key piece but how much do we need to know in order to to do different things and just how much of this is um perhaps from like uh just just background. Even though it's hard to say what what's the uh background and what's the key piece because there clearly this chapter was written to to try to at least give a uh some balanced but minimum uh blend of of true background topics like fundamental properties of probability distributions logarithms and derivatives [Music] um probabilistic models probabilistic models that have reference with about observations and actions. So we can continue going on, but I think like th

those kinds of statistical models could be applied and useful w with or without understanding or being able to to know all of this. Um, however, it's it's good and and kind needs to be at least understandable how the tools and the implementations relate to what their formal structures are. What what what does anyone else think about it though? because definitely that wave kind of hits hard in chapter 4 with with just just being kind of half equation-y. I I don't know. My my my personal feeling is that the the equations and the sort of computer modeling seem to be kind of disconnected from each other in practice. I just get that feeling. Do you feel that way for like using linear regression models like just using statistics and papers the way that P values are used? I'm I'm just curious. That's kind of just where I calibrate the question to. Uh no, not not in the same way because um all these equations are supposed to like represent things in thermodynamics and you know I don't see any uh thermodynamics in like the computer models. Yeah, I think you know this is something that that's um a uh an interesting ambiguity or sort of juxtaposition or like a a namespace crowding with like energy based models. So there's like energy as as the potential and the kinetic energy of of some object and mass for example. um and and this sort of sense of using an energy function to to fit some statistical model. And then there's like if, whether, how that actually connects to thermodynamics like candles burning and um the energy density and like temperature dynamics, but it's possible to make these energy based models that that but it's not talking about like energy in the gasoline burning, but that's kind of like they they they are in different domains, right? Yeah. Well, so so I understand that the word energy and AI can mean all kinds of things. So even in math, like for example, there's something called the deer energy, which you know, but um I I my understanding was that this was supposed to tie into thermodynamics and equations were supposed to be interpretable as thermodynamics. Yeah, I there's there's um there's a few kind of separable angles on that. So if you follow the basian mechanics um like Dalton Saka Devel and um that line of work then it's like generalization of thermodynamics just in the same uh set as the Newtonian the thermoinformational and the quantum mechanics basian mechanics being like um describing those belief distributional updatings and other so that's one sense but that doesn't mean that it has to connect to uh thermo in terms of describing the energetics of systems it still could just be an you could just be talking about some Gaussian distribution doing some rotation it doesn't have to like tie to the actual survival of some embodied object But then there's that that that second angle is like is this relating to the actual thermodynamics of systems and and how they maintain themselves above and beyond what um just not dissipating which is like the kind of autooosis. That's just sort of how how

even qualitatively the biological systems are framed. So there's but but that's a little bit of a separate um question almost or or area of of applying it than just trying to generalize the theoretical physics of information systems. Daniel, if I could jump in real quick. Uh I really appreciate the way you bring so many disparate fields together. Uh and I also appreciated Sylvia's question. Um there's so many levels to talk about this, but I think it's really important always to circle back to the principle because that's different than the math. The the principle of free energy. um at being a framework. This is where I've come to um an understanding uh that it's really important to understand the framework. A it is a framework that uh is applicable in so many different ways. uh and I had to go back and look up Elmholtzian free energy and uh you know the the the chemical free energy all in terms of thermodynamics thermodynamics they uh can be contextualized under different conditions but really the principle provides a framework to apply so many different um fields which I think is the brilliance of using this framework. It allows using a structured set of equations that really uh you could go into with a lot of depth, but you have to understand the language, the what the symbols mean. I think this was sort of getting at Sylvia's question. uh you have to understand what the terms are, what how to construct the equations in order to u be able to construct particular applications. But there are so many tools to use um uh in in terms of what John was talking about you know the computer modeling is more about classification uh whereas the um mathematical equations are just giving the rigor uh uh to to the context. Um I uh I I I want to hear you talk about this more Daniel but um uh bringing in those who want to apply the principle to these different applications whether it's on the molecular level or the social level uh that the the framework can be used. I I I really think that's the value of the free energy principle is it's just a a a common language that can be used in all these different applications and if u even if a person doesn't know what the symbols of the math means you know they in consultation with others they can find that out but just the understanding that it is a consistent universally applicable framework of how systems evolve in this free energy principle, no matter how complex it is. I I don't know if that's helpful, but this is just my approach to contextualizing all these different layers, these different applications. Uh it's centered around this principle. Um so on Helmholtz free energy so it it looks like that the sort of fristen free energy is um you know supposed to be the same as in addition to other things the helmholds free energy because because of that uh energy and entropy term where they talk about it being proportional to to energy the proportionality of course being temperature but I so the the weird thing is that that energy term comes from the Boltzman distribution. And so I'm wondering if there's a more

general form of the equation that can deal with other distributions or and in particular one thing I worry about is uh the Boltzman distribution is one one right but if you have some distribution that's not one one say a Gaussian uh then it can be then trying to compute the energy from the probability isn't going to work because you don't have a one one function. I I'm just questions like that. Yeah, lot lot to say just to these last points. This makes me think of this uh one paper which I'll put in there like what makes something what makes an equation an energy function? I I'm I'm I'm actually just at some there's probably lots of levels to answer this. What makes something an function? What makes it a free energy function? And then like what what are all these different flavors with adjectives or with last names like oh this is this approximation. How many layers deep into qualifying like what kind of it it is matters for the FEP and h how many how many flavors of that matter for applying like that does it are there in practice across different settings um or or is it going to is it going to get super tied down in the weeds with needing to specify all these really different settings for for um and like slightly different uh unjustifiable assumptions will lead to very different approaches with different um different like computer approximations or or will like uh just multiple full pretty uh different uh approaches give give broadly similar results. So for example in the SPM uh textbook the there are chapters that have elements with classical parametric statistics basian statistics and nonparametric statistics. So like the bootstrap and shuffling and relabeling techniques like procedural techniques to randomization and these and the classical and the basian form of statistics and I and and probably things like conformal or other statistics too. It's like if um there really is a warmer room and a cooler room and the thermometer has nonzero information, you pretty much should be able to detect it. somehow. So that's the that's the kind of empiricist part and I think that and and this is a perfect chapter 4 question because it's like okay low road was just about making the computational building and assembling the joint distributions for basian models. High road is more theory and axiom driven. So then it's like okay so how um how general or um or or relevant is is this meeting point? Is this like every single model like even just a linear regression? You could interpret that as as doing some process describable like this. Like does this just redescribe and generalize statistical methods or or or what what what is really being said or claimed beyond what Well, you you asked the the precise critical question. What are the important constraints in constructing your model, one's model? What are the important variables or or or factors to to consider like are is the noise randomly distributed and independent or not? you know that determines what kind of statistics you're going to use. You know, is it is it a constant pressure system for Gibsonian free

energy or is it a constant temperature system for uh what is it? Hamiltonian or hem helmholtz you know it's uh could you consider the activity of neurons in the brain as a uh more or less uh constant energy but redistributed um it it it really is a creative process in figuring out how to construct the problem and I I think that's why uh it's helpful to understand here's the framework. Here are the tools to use. Now, how can we rearrange the problem uh to make the appropriate assumptions uh to use the formula in the proper way and that's going to take a lot of I think collaboration or no individual brilliance but uh collaboration would be more helpful to me. Yeah, those are great comments and it reminds me a lot of complexity science and system science like also that is about a framework and approaches about patterns and systems of different scales. um and also has long featured agent-based modeling like using net logo um and other agent-based modeling approaches. So the that that is a big part of it is um and then where where do the so just beyond that qualitative structural um possibility of complexity science? What do we really get with chapter 4? But it helps connect all the way to a statistical like looking at um neuroimaging and SPM. And so thinking about first in uh prefab working with just this one defined data input type like fMRI data um or or uh EEG data. And so kind of constraining on what the the input data were and then seeking to make cognitive inferences about perception and action. So that and then um it it it kind of built many of the empirical precursor threads like fitting latent spa fitting in uh fitting empirical data noisy empirical data uh finitely. So that's kind of the connection point with the with the empirical nature of of data in in like research um with projecting onto latent spaces whether just generalized dynamical systems latent spaces or or information spaces like probability distribution spaces which uh is a space where there can be more axiomatic constraints on like thingness which in many situations can be kind of like a just a tautology or or an auxiliary point or overkill related to just statistics and practice. But it gives a lot of touch points with some of these statistical information underpinnings which then tie into at the very least like the efficient ways to model and identify systems empirically and then I think there's like a lot of going further than that to this like against that second angle kind of like to John's question isn't this about the thermodynamics it's like but isn't this really about information processing not just doing um base optimal statistical modeling of empirical data like Chris Fields's course wasn't about saying here's how you fit the models that balance the right number of parameters with the right not overfitting the data where like complexity minus accuracy is just a normal measure for not overfitting a statistical model, but that doesn't need High Road. Uh, or it doesn't need a 4. Here's the question. How would you determine the actual free energy of say I I have a cat here.

How do you determine the free energy of my cat? It's like what's the what's the mean squared error of your cat? It's like a model specific. If you're if you're looking for a number, it's just a feature of we we can look at equation 2.5, but it's not like it's defined outside of a given distribution as models, right? Well, so one issue is that that the distribution is the Boltzman distribution, right? Because you you can tell that from the energy equation. Um but the Boltzman distribution assumes that you have a closed system and that the uh the energy is constant. But a cat is an open system. His energy can change when he runs around or eats. Um so like I this is this is a thing my physicist Fred brought up to me is how do you determine how do you act how do you actually use this on a complex system? Uh John, real quick, that is exactly the question that I ask so often. How do you set the bounds of the system? Uh clearly at ultimately you're talking about the universe, but if you're just talking about the cat, you have to make certain assumptions for the moment, you know, for as an uh and I know you like complex systems also but you can uh set the bounds of the complex system even as as temporary approximation but but that those kinds of approximations will only work if things are of the right scale. So uh if the energy is changing fast enough then you it's not it's not going to be approximated by a Boltzman distribution. Right. Yes. And then you have the hierarchical levels. I think this is also brought up not in this chapter but uh you just layer and layer and layer your models. Exactly. Great point. So there there's probably a lot in what you said there um John and like right the identifying the right scale. So one sort of uh practical note on this is like it may just be constrained by the data that's being obtained like it's just fMRI data. So it's it's constrained versus the more like open multiscale inquiry and then just like like um Stephen said like this is where structural modeling and and nested systems but also there there's that kind of Bucky Fuller like starting with universe idea and kind of be and then discarding irrelevancies and like the subtractive method to identifying systems boundaries. Um and then just in terms of model identification that's where there you do uh structural ju just like you could do with a multi with clinical data like they just make nested the the model comparison amongst um every single variable every joint and then just ask does this model with more parameters explain more? Does it have more accuracy minus complexity g you know given explaining the variance in this data set given the number of parameters and then so but treating that as a model selection model inference task like model selection as inference and then identifying so okay so we have um we have data coming in on a continuous spectrum And then it's and then so compare how many different ways to um partition that. Should there just be one Gaussian? Should it be a mixture of Gaussian? Should it be mixture

of nested Gaussians? But in a given situation that might be like overexploring, but some of these are pretty well typed out at least in theory. So being able to do at least kind of broad sweeping explore a few structurally different hypotheses pretty fast and and and it's a super rich question. There's a lot to explore in theory, but in practice it might just be easy to identify. But it's pretty fascinating that there's also a lot to explore in that question about like the general systems theory and nested systems in their in their general um interactions and and that may lead to some interesting new possibilities in real world systems. But in terms of of like what data is available be like we have um 10 replicates per person from three conditions. It's like well there's that's that's the hierarchical level of the statistical model. So that's just more just making statistical models that recapitulate the act the levels of identifiable structure in the data. I I'd like to point something else out which is the the the the the combination of in the framework the a generative model and um uh using picking an appropriate general generative model within this framework. And just to point out in the brain there are different generative models being used like in the visual system there's this convolutional network and in the uh you know reward system basal ganglia cortex you've got this reinforcement learning generative model that they're different types of deep learning models but they can still be but they're still within if in in our world they're within this free energy, you know, basian type most probable uh selection process. I I just I guess my question can be phrased as um if something I if something if if my cat say wasn't acting to to minimize his um to minimize his fist and free energy, how would we know? like what what test could we do to find out uh whether he is or not? And again, it's it's not really a question of in my view exactly that um what is um what is happening but rather how can we uh formulate it? How can we describe it? And really all free energy does is provide an upper bound uh you know it's it's like um it's it's accurate enough for the task. It's complex enough for the task it in terms of how we describe it. We can make predictions about what's happening. Um uh we can model this system as again using the brain as a reference. Um there's a a a system of actors that are reacting to what's happening in the environment. And the way that they're formulated is in in response to what's happening in the environment, there are created these priors and uh it in uh behavioral uh interaction with the environment. There are these uh predictions of what's going to happen. uh we say predictions but it's really just the way the system is set up in in a developmental response to how it has been interacting with the environment all along. You know nothing springs out of nothing. It's it's a developmental process and it's this this is how the uh the the free energy formulation is a great descriptor of that process. It's a great framework for thinking

about how that process occurs. Um, uh, I get that you make models and then you use the models to make the predictions, but if you if you want to make a model of my cat VV, its free energy, you're going to have to try to figure out what his free energy is. And that brings me back to my previous question of how do you figure out the free energy of a cat? Well, what's your model of a cat, I think, is the Um, I would like to I I would like to be able to uh predict the cat in the future. It's what you said. You you have predict to predict just the body temperature or blood sugar or the location or cognitive let's say location because that that that one's complex enough that it it bring it brings up the problem. How like Yeah. How do I know where he's going to run to? So that's probably doable actually. Yeah. So you have a location in two dimensions in the space. There's some distribution over X and over Y or there's some probability kernel summing to one spread out over the the the zone that you're going to be modeling this. So then it's like there's some diffusion process or there's some other underlying stochastic process giving rise um to samples coming from some true underlying latent distribution. And then it's like if you're using GPS or a video or just spot sighting, you're getting some empirical observations of the location of that cat. But but what if the so but if you do that like what if the variance is too high to do anything useful with it? Yeah. Big picture the whole point of statistics as a framework is so we can differentiate between like where do we have a super tight Gaussian and where do we have one that's so washed out that it's just like a flat line like it has infinite variance or something like that. So big picture that's the whole point. be able to do estimates of variance like hey we haven't sampled enough to reduce our uncertainty and that doesn't mean that with some other prior or some other framework or or or framing there isn't information on data set it just like those observations hitting that distribution resulted in a wash out which could be a false uncertainty so it's not that any model is going to be like etc no free lunch but that that is the doing of statistics is determining the information content and and the relevance of the data on updating your prior and understanding whether you are just getting your prior back um or whether you're actually picking up with variable starting positions in prior families similar tendencies in in data. But for something like a cat in order to get a a variance that would be low enough to actually do something with you have to like model like the whole brain of the cat. And this would bring into bring in issues of how do you estimate the cat's free energy, right? No, you might just have a a a a spatial temporal kernel learnable through any number of means including just all kinds of methods. Deep learning, parametric statistics. There might be so many ways to reduce your uncertainty by many bits about where the cat's likely to be. You can make a transition

matrix by room and maybe just there's a circadian cycle. Now where that exists between providing no information like it's its behavior is encrypted like maximum or whether it's super recurrent with a high information learnable content. That's just an empirical question. Well there's no there's no there's no ability in modeling any given situation or whether there's a given signal for you in any given data set that that it it that it's just it doesn't take down statistics as a framework or bypass it. It just means there are situations that are with high identifiability for certain modelers or not. Well, well, I think the idea that his behavior is encrypted is sort of accurate because animals are supposed to be unpredictable to other animals. And so for that reason, I think you you would have a very high variance and you wouldn't actually be able to model it unless you were able to model all the sub aspects at the lower scales. Uh and then that sounds really really hard. Well, John, uh, it needs to be unpredictable enough. It it needs to be also conversely predictable enough. Like there may be vast stretches of time where they really don't want to be predictable, but then there are other local stretches of time where they want to be very predictable, like when the food is going to be there. So, there are going to be uh uh again, you don't have to know everything to make the model. uh they're going to be black boxes. Well, well, so in chaos theory, if you want to model something linearly in time, you have to have exponentially more information. Again, this was Daniel's point. It depends on how, you know, uh how fine grained you need to be. Yeah. Let's say I want to know what square foot he's going to be in for the rest of the day. Yeah. Yeah. Like what what performance you get with what input data and what spatial grid and and what kind of reliability you need in the statistical model. the that those are the exact questions that are important for statistics um in practice like data engineers um behavioral researchers like doing statistics and determining I'm probably all other kinds of data analysis questions but this this is the meta science of the approach to to have an open-endedness structurally in these settings and and uh just practically to explain data. Well, and and in practice to use a a multitude of statistical and non-statistical methods and also uh sampling methods. Like if you want to know the square foot location of a cat, you're going to have to at least sample, you know, uh half square foot resolution of the entire space, you know, the closed system. Well, I I think you'd have to sample more than just location. I think you'd have to do all sorts of sampling about about the cat's nervous system as well to to get the variance anywhere near low enough to use for anything. So that that that's again an empirical question. Are you going to have is your data just is your model constrained to just sampled location data in latent location data inferred you know and/or are you opening up this structural possibility of well

what if we do latent state inferences like what if we also have on this other side information do we want to include temp temperature in the room or do we want to include the season like what other variable should we include how should we mix and match and compose and and flow these variables the all the hit it hitting it with the all by all statistical analysis which it might be doable in certain settings but it's it you're right that it scales computationally poorly whereas for interpretability and efficient computing the sort of um goal approach is a sparse model where where the sparsity corresponds to like the sort of cutting nature at the joints. doing actual relevant identification of ontologically real systems. you know, caveat, you know, what is real, what is useful, but just aiming somewhere in that useful direction and in a less and less uh predisposed more and more structurally flexible way do multiscale identification knowing that analytical systems identification a certain way would be implementable. just with certain just you know knowing that there are computational tools to implement the models beyond just sort of well you know let this variable represent this and let this represent this and kind of just purely going off into a simple direction. I I'm I'm just wondering if the free energy principle can be tested on animals or if it's just not computationally going to be possible to do that. Like does applying it to choice behavior counts or are you talking about like the the what the textbook calls the the whole iguana? It's like I don't know. I I I I guess I guess I would be happy with seeing something impressive that couldn't be do done by another means of analysis. Well, if I may, this is it's also applied to social systems. I mean, you know, it's there are several papers on uh psychiatric uh uh applications and it seems to be pretty successful. I think, correct me if I'm wrong, Daniel, but I think baked into this approach is the probability of the uh how important you know the probability that unknown variables are important or not. uh um you may not know all the var you may not be able to model all the variables but well that's just sort of the variance right well yes but there are different kinds of variances uh and and I well yes different kinds of variances you're right that's that's what we're talking about that are being um iteratively uh adjusted uh the you know, updating the priors in order to get a more accurate probability spread, you know, uh, complexity, the the variational probability. I just wanted to jump in as as a novice here and you mentioned the priors and you also mentioned other approaches. I'm wondering if this is um I guess it would be an application of of free energy as opposed to another approach if you're doing updates um based on uh new observations that generate prediction errors. But I I'm thinking looking at my cats um how how much do you weigh just priors? I mean, if if you were to just ask me, hey, um, you know, as as a relatively lay person in in this discussion, like where do you think your cats

are going to be in in a given time? I say, well, what time of day is it? This is where they are. This is where the sun is. And so, like, a lot of the variables seem seem kind of predictable. So, then you've got this really strong prior and your generative model is probably fairly predictive. And then I guess the next thing is well then how do you add to the complexity of some of the unknown variables? Uh there's a loud sound outside. Uh you know, things that are unexpected, right? I'm I'm looking Yeah, that was that was a great point. And it's kind of like the figure 42 thinking graphically and what is that Lego toolkit of graphical assembly diagrammatic assembly of uh observables and unobservables. So it's like that's that's and then which and that kind of gets into abductive processes. Um model selection, structure modeling, selection amongst statistical models but mo but especially once you can create unobservables or do latent state inferences you did I mean arbitrary chains of latent states that that can be transformations or interleavings or just different patterns like one pattern that that people used to model the brain like was there there's convolutional like patterns structurally um there are hierarchical predictive processing like uh functions this is kind of like Mars level of analysis like what are the the algorithmic and the cognitive and etc layers or what what is happening And then how is the structural the real dendrites and and ga and everything. How is that structural um mechanistic layer for for different embodied systems or a silicon chip or whatever related to the symbolic diagrammatic relationships of of um observables and unobservables and different transformations. Yes. Uh, John, you you you brought up a philosophical question. Actually, you may think you're wanting to model your cat, but there's no such thing as a cat. there's a population and uh you know a cat is a member of the population but it doesn't exist independently. I mean, again, I'm talking philosophically here, really. Uh, you want to model everything about your cat, but you're absolutely right. To model everything about your cat, you'd have to model the entire universe or at least the entire, you know, world that that cat inhabits every place, you know, including other cats and, you know, everything interacts with. But but but what I'm worried about is like what if he what if he's not lowering what if he's not acting to lower his free energy right now? How would we know? Um that that's my fundamental question is uh maybe maybe he's acting to lower his free energy and maybe he is not. How do we know? Well, isn't free energy just an upper bound to uh the difference between the complexity and the accuracy or the the divergence and the evidence. You know, it's it's it's not a precise number. It's just an upper bound to what the uh difference between the energy and the entropy is. Yeah. Look, that's that's one great point. Let me give another piece on this. Here's in in um chapter six complete class theorems state that

any statistical decision procedure so whether it's um ruminative repetitive maladaptive you know um just leading to demise leading to peril externalities failure may be framed as base optimal under the right set of prior beliefs. So the whatever whether the cat is staying in one room or whether it's between two rooms or whether it what or whether it leads to its demise that can be toddloically modeled as its own as a distribution. So there's no there's no sort of perceived or or projected pathology on like human or a cat system or any behavioral situation you can dream of that that couldn't be seen from a modeler's perspective from the outside as a statistical decision procedure. Okay. So are you familiar with David Deutsche's criterion of good explanations? No. So, a good explanation has to have something called difficulty to vary. Uh, which means you you you can't just change all of the details. Uh- which means that if you're you know, if you had a negative experimental result, you can't just change the details to get another one. Um, like I if the if the idea is you just go through the the enormously large space of all statistical models until you find one that says the cat is minimizing its free energy. Un unless there's some concrete way to actually check whether that statistical distribution applies to the cat in some way that there there you're not going to have difficulty to vary with an explanation like that. Yeah, it that's it's it's what if you think it is it possible to talk about the statistician reducing their mean squared error for their predictive model cat? Like do you extend that sort of belief to all of statistical modeling and just doing behavioral science? Because that's that's a separate avenue than asking what the cat is really doing and taking this sort of like ontological like what is the what is the territory really doing is a little bit of of a of a extension to the the the systems identification question. And so just it's it's just what what it's they're a little bit different application questions. It's just that it's just that cats are sneaky and they're temperamental. So they may they may decide that they're not going to lower their free energy because they want to confound us. And so how would we know that they're not? So there's a ton of ways to think about it, but just like from a complex system like or system science, general systems perspective, I just think there any particular data point you throw, cats are sneaky or this I think this will happen or they're like this or I saw this or I don't think that's possible. It's like we're looking for something that includes all of those expressions, but just like but then this this mushroom did this next to this tree, so how could that be doing that? It just those are all inquiry directions, but the whole point is what is the grammar of all of those identifications and assertions that you're just saying in a row. there's a bigger. So, it's nothing nothing about cats specifically. It's just what is the language that you're looking for design space within with those kinds of claims you're making about your cat. Well, it's sort of

where does the probability distribution come from and what what does basian inference add that wasn't there before in trying to figure out the probability distribution? Yeah, fair enough. I think you're asking really good questions and I you try it and see if it works. Yeah, that's a great way to put it. And also and I also think about like are we talking about modeling a situation relative to just vibing and not modeling a situation or what what or are we talking about the difference between basian statistics and parametric statistics and non-parametric statistics and deep learning or is it methods comparison within we're already in a probabilistic modeling technical um sort of setting or is it a philosophical qualitative discussion about the nature of things that isn't really looking to be just deflated or resolved to like yeah what is the free energy of marathoners at this mile it's like well are we talking about the temperature okay well here's the data set here's the value it was 2700 okay let me let me put it this way so so my question can be phrased like this if if I'm trying to model something by introducing some kind of probability distribution and using that to model it. What do I get out of using basian mechanics that makes that better than it would have been without? So, you know, anyone could just start from first principles and try to model things with a probability distribution. How does that modeling process change if if you're using the free energy principle? Yeah, it's a great question and I I look towards more and more really cool documented more explainer paths through that and so beyond just looking at the literature and the citations for what are described in this textbook. Um one I'll just give two two two short ways. One way is like as a perspective and a a stance. This is like how information is like using this as an as as part of our toolkit of information theory and understanding statistics. It's like this is just how statistics inference is. If your joint distribution that you're looking at is just like just it it's a lens on what statistics have already occurred. you just look at the statistics in a a clinical study or behavioral study. So in that lens it's just a way we can think about even those statistical analyses. But then as we start thinking more into um basian graphical models, it it gives composable modeling tools and then it just it you you can do statistics modeling and have these diagrammatic tools. And there's other situations where it's it's convenient and more than adequate to do um like to think more about a data science flowchart like here we're just going to push it through a neural network and get the output. So that's thinking like at a systems uh engineering level and just letting a ton parameters with a ton of uh models with a ton of parameters just learn relationships and data. Here's an explicit probabilistic distributional approach that has some other features and and those features are it plays well with other techniques, other models. it it allows interaction um translation between different

approaches. Uh you can start with a a very coarse grain approximation and update it um to make predictions in a way that can be interactive with other approaches. is my view of this is the advantage of using this approach. It it gives you more tools. Yeah, I think that compatibility of free energy with other, you know, I guess you could say simpler models is is appealing and and also it seems that if you reduced it to its simplest application, it also holds up. So in other words, there's there isn't something that's too simple um to be addressed with with you know active inference. That's a great point that with with simple dynamical systems we can get into the trivial and the inert like the function y equals 5. It's like, well, that's overkill to have a function to have just this line going for for infinite time at five when we could just say the number five. It's like, yeah, but then it can it's kind of like a it has all these other features. So there's there's there are statistical models that would be um useless or overkill or underkill. That's the whole model selection question in theory and practice. But in terms of the um latent space structuring, it can be um a coin flip world or it can be all this structured latent spaces and and and structural models with different state spaces, different time concepts. Can I suggest that that's actually a criticism of active inference in the free energy principle that it is so available uh to u interpretive description that it can describe anything depending on how you set up the equations and the models. I mean its strength can also be a criticism of it in a way uh because you you really have to um trust I guess how the models are set up and again always proof is in the pudding so to speak that the the you does it work um I I have heard people say that it's Um, I don't understand it. It's it's it's like it you can arrange the components of the equations in a way that it's sort of like h having uh any model with enough parameters can describe any anything. Um but the advantage is it's predictive uh and and uh and flexible enough to bring together different models into one system. Yeah, this is one funny math like quote to to the that point. Um, with four parameters I can fit an elephant and with five I can make him wiggle his trunk. So it's like um w with modeling it's just really what is the what is the role what is the person's expectation of the role of any of this of the low road or the high road? How much work do people expect or what kind of foundational grounding do people expect in statistics? What do people think is under the hood of a t test or a p value uh or a baze factor? Do people know about the the zcore and the p p value factor and they're looking for other criterion for signal processing and empirical systems identification or are we pulling back a step like what is the role or the function of any quantitative measure in comparison with just choosing not to do Yeah. So if I have a a patient in the clinic who can't walk straight uh and I build a model that says that there's a part of the

brain that predicts uh what forces are necessary on certain muscles to get to walk straight. And there's another part of the brain that's interpreting the sensory input that's uh letting the person know if they're walking straight or not. And I set up a model that would predict there would be an error in one component of this system. There would be a a a damage to one component of the system. I'm going to now go look in this person and see if there is a part of their brain or nervous system that's not functioning properly or there's damage to it. And voila, there is damage to this part of the brain. It that that's a use a useful use of that predictive modeling. Yeah. Well, a lot of interesting uh comments and and aspects today. So, thanks for joining. You could uh come next week at the same time discussing chapter 9 i.e. on uh empirical data fitting. So that kind of relates to what we discussed a little bit today or see you in two weeks. So thank you. Bye.